02. FIRST JAVA PROGRAM.

Structure of java file.

* Every thing written in .java file must be in classes as we can say that every file having .java extention in a class.

* A class with same name as file name must be present in .java file.

* First alphabet of class name can be in upper-case. It is the naming convention of class name. however, it is not compulsory to do so.

* class which is having same name as file must be public class.

* A main function | method must be present in the public class, main is a function from where the program starts.

Converting .java to .class

* Using javac Compiler we can Convert .java file to .class.

Command to Convert .java to .class

* `.javac and .java file name`

* Let the name of .java file is Main, So the Command to Convert .java to .class is

* `javac Main.java`

* Above the Command Create a .class file (Main.class) which Contains Bytecode.

Running the program

* By using java and name of file we can run the program.

* `Command : java Main.`

Hello world program:

```
Public class main {
    Public static void main (String[] args) {
        System.out.print (" Hello world ");
    }
}
```

output:
Hello world..

01. Public (in first line): public is an access modifier which allows to access the class from anywhere.
02. Class: It is a name group of properties and functions.
03. main: It is just name of class as same as the name of file.
04. Public (in second file): It is used to allow the program to use main function anywhere.
05. Static: It is a keyword used when which helps the main method to run without using objects.

06. void : It is a keyword used when we do not want to return anything from a function / method.
07. main : It is the name of method.
08. String[] args : It is a Common line argument of String type array.
09. System : It is a final class defined in java.lang Package.
10. out : It is a variable of PrintStream type which is Public and Static member filed of the System class.
11. println : It is a method of PrintStream class. It Prints the arguments passed to it and adds a new line. Print can also be used here but it prints only arguments passed to it. It do not adds a new line.

What is Package ?

* It is just a folder in which java files lies.

* It is used to provide some rules and stuff to our Programs.

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Primitive data types.

* Primitive data types are those data type which is not breakable.

  * Ex: String is not primitive data type so we can break this data type into char

    * i.e, String "Sharath" can be devided into 'S' 'h' 'a' 'r' 'a' 't' 'h'.

* But primitive data types are not breakable. We can not break a char, int ect.

List of Primitive Data types.

01. Int (Integer) : It is used to store numeric digits
   * ex: int i = 26;

02. char (characters) : char is used to store characters
   * ex: char c = 'A';

03. float : float is used store floating numbers
   * ex: float f = 98.67f;

04. double : double is used to store large decimal numbers
   * ex: double d = 676767.676767;

05. long : long is used to store numeric digits which is not able to store in int
   * ex: long l = 67676767676767

06. boolean : It only stores store t or f values. i.e, true or false
   * ex: boolean b = false;

* Important tip:

* In float and long we have used f and l, if denotes the number in the variable is float or long type, if we do not use this java considers float value as double and long as int.

* Literals : It is a synthetic representation of boolean, character, string, & numeric data.

* ex: int a = 10;
Here 10 is called literal.

* Identifiers: name of variable, methods, class, Packages, etc. are known as identifiers.

* ex: int a = 10;
Here a is identifier.

Comments in java.

* Comments are in java something which is written in source code but ignored by compiler.

* Two types of Comments.

01. Single line Comments: Used to Comment down a single line (// is used for it).

02. Multi line Comment: Used to Comment down multiple lines (/* */ is used for it).

Inputs in java.

* We have Scanner class available in java.util Package to take input.

* To use this class we have to

01. Import java.util.Scanner in our file.
02. Create a object of the Scanner class.
03. Use the object to take input from the keyboard.

Syntax.

```
import java.util.Scanner;
Public class Main {
Public Static void main(String[] args){
   Scanner Sc = new Scanner (System.in);

}
}
```

01. Scanner : It is a class requiured to take input, it is present in java.util package.

02. Input : It is an object that we are creating to take input.

03. new : It is a keyword used to create an object in java.

04. System.in : System is a class and in is a variable that denotes we are taking input from Standard Input System (i.e keyboard).

* int input : nextInt(); is a function used to take input of int.

* Syntax.

```
Scanner Sc = new Scanner (System.in);
int rollno = Sc.nextInt();
```

* Float input : nextFnt(); is a function used to take input of float.

Syntax

```
Scanner Sc= new Scanner(System.in);
Float marks = Sc.nextFnt();
```

* String input : Two ways to take String input.

01. Using next(); method : It will take one w input till a Space occurs.

02. Using nextline(); method : It will take all Str input including Space.

Syntax.

* method 01. : next();

```
Scanner Sc = new Scanner (System.in);
String S1 = Sc.next();
```

Input

Hey Sharath.

output

Sharath.

* method 02 : nextLine();

```
Scanner Sc = new Scanner (System.in);
String S1 = Sc.nextLine();
```

Input

Hey Sharath

output

Hey Sharath.

Type Conversion

* When one type of data is assigned to another type of variable an automatic type conversion will take place under some condition.

Conditions

01. Two types should be compatible
02. Destination type should be greater than the source type.

Type Casting

* When we convert one type of data to another type is known as type casting

e.g. Ex: int num = (int) (67.67f);

Automatic type promotion in expression.

* While evaluating expressions the intermediate value may exceed the range of operands and hence the expression value will be promoted.

There are ~~two~~ some conditions for type promotion.

01. Java automatically promotes each byte, short or char operands to int when evaluating an expression.

02. If one operand is a long, float or double the whole expression is promoted to long, float or double respectively.

```
Ex:
    byte a = 40;
    byte b = 50;
    byte c = 100;
    int d = (a*b)/c;
    System.out.println(d);
```

// Here when a*b occured it became 2000
// which is out of range of byte so here byte is
// automatically promoted to int type.